

DILEEP BODDU

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[GitHub](#) | [Portfolio](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

AI/ML-focused BCA (Data Science) graduate with hands-on project experience in machine learning, deep learning, computer vision, NLP, and data analysis. Built end-to-end solutions for AQI prediction and image caption generation, covering data preprocessing, model development, evaluation, and application development. Seeking an entry-level AI/ML Engineer opportunity to apply practical machine learning skills to real-world problems.

EDUCATION

Andhra University – Gayatri Vidya Parishad College <i>Bachelor of Computer Applications (Data Science) – 8.81/10</i>	2022 – 2025
Sri Vasavi Junior College <i>Board of Intermediate Education AP – 9.18/10</i>	2017 – 2019
Vijayam School <i>CBSE – 10/10</i>	2017

PROJECTS

Air Quality Index Prediction using Machine Learning <i>Python, Pandas, NumPy, Scikit-learn, XGBoost, Random Forest, Matplotlib, Seaborn, PyQt5</i>	2025
- Built machine learning models to predict Air Quality Index (AQI) using PM2.5, PM10, SO2, NO2, O3 and NH3 as pollutant features. - Implemented and compared XGBoost and Random Forest regression approaches; XGBoost achieved an R^2 of 0.81 and Random Forest achieved an RMSE of 14. - Performed data collection, preprocessing, model training, testing, evaluation and visualization using Python, Pandas, NumPy, Matplotlib and Seaborn. - Developed a PyQt5-based application interface for entering pollutant values and obtaining AQI predictions.	
Image Caption Generator using Deep Learning <i>Python, TensorFlow, Keras, ResNet, CNN, LSTM, NLP, Flickr8k</i>	2025
- Implemented an encoder-decoder architecture where ResNet acts as the CNN encoder to extract meaningful visual features from input images. - Developed an LSTM-based decoder to generate a sequence of words describing the visual content of an image. - Handled image preprocessing, Flickr8k dataset preparation, text tokenization, sequence preparation, model training and caption generation. - Evaluated generated captions using BLEU score and achieved a BLEU score of 0.64.	
Personal Portfolio Website <i>React.js, HTML5, CSS3, JavaScript, Responsive Design, GitHub Pages</i>	2025
- Developed a personal portfolio website from scratch using React.js to showcase technical projects, skills and professional information. - Implemented responsive design for desktop and mobile devices and managed version control and deployment using Git and GitHub Pages.	

INTERNSHIP

Python Intern — DataPro Institution	May 2024 – July 2024
- Applied Python programming concepts through hands-on exercises and small project-based tasks during the internship. - Worked on practical problem-solving and application-oriented Python development activities.	

TECHNICAL SKILLS

Programming: Python, C, R, SQL, Java
Machine Learning & AI: Machine Learning, Deep Learning, CNN, LSTM, NLP, XGBoost, Random Forest, Scikit-learn
Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Tableau, Power BI, SPSS
Frameworks & Web: TensorFlow, Keras, PyTorch, React.js, JavaScript, HTML5, CSS3, PyQt5
Tools: Git, GitHub, Jupyter

CERTIFICATIONS

- Programming for Everybody (Python) — University of Michigan
- Python for Data Science — IBM Developer Skills Network
- Machine Learning Certificate — Reliance Foundation | Skilling Academy
- Internship Certification — Python, DataPro Institution
- IoT Workshop — Teckybot
- 3-D Printing Workshop — Teckybot

ACHIEVEMENTS

- Coordinated a campus wellness initiative for 500 participants, supporting student engagement and community participation.
- Facilitated a Clean Swachh Bharat workshop for 200 college students, promoting environmental awareness and sustainability.
- Conducted community surveys and documentation for a 2024 project analyzing the environmental effects of plastic pollution.